TAAP: Timestep-Adaptive Attribute Preservation for Identity-Preserving Face Generation

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Abstract

Standard diffusion models generate high-quality images but struggle with preserving semantic attributes. TAAP (Timestep-Adaptive Attribute Preservation) addresses this by learning per-timestep identity loss weights through a lightweight network, integrating seamlessly into training. Experiments show that TAAP reduces FID by 21% over vanilla DDPM and outperforms fixed-weight baselines, demonstrating that adaptive weighting is superior for attribute preservation.

1 Introduction

Denoising Diffusion Probabilistic Models (DDPMs) have emerged as powerful generative models [4], offering stable training and high-quality image synthesis compared to GANs [1]. However, vanilla DDPMs often struggle to maintain semantic attributes, particularly facial identity consistency, and thus frequently generate visually plausible yet identity-inconsistent faces.

A promising direction to address this limitation is to incorporate identity-preserving losses during training. Recent works demonstrate the effectiveness of such strategies: some apply explicit identity conditioning with uniform timestep weighting [10], others introduce identity adapters that enable zero-shot identity preservation [12], and others adjust the likelihood formulation to maintain identity while allowing diversity [7]. Collectively, these methods show that identity constraints are a viable and effective strategy for controllable face generation in diffusion models.

A straightforward baseline is to apply a fixed, constant identity loss weight across all diffusion timesteps. However, this uniform weighting is fundamentally suboptimal because it assumes all timesteps play an equal role in preserving identity, despite the heterogeneous nature of the denoising process. Early (high-noise) and late (low-noise) timesteps likely contribute differently to identity retention. In my experiments, this fixed-weight baseline surprisingly performs worse than vanilla DDPM, suggesting that indiscriminate identity supervision can harm overall generation quality. This motivates the question: can identity supervision be better aligned with the timestep-dependent structure of diffusion?

To this end, I propose **TAAP (Timestep-Adaptive Attribute Preservation)**, which learns per-timestep identity loss weights via a lightweight auxiliary network. Instead of the standard fixed-weight formulation

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DDPM}} + \lambda \cdot \mathcal{L}_{\text{identity}} \tag{1}$$

with a timestep-dependent weighting scheme

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DDPM}} + \alpha_t \cdot \mathcal{L}_{\text{identity}}, \tag{2}$$

where $\alpha_t \in [0, 1]$ is a learned importance weight for each diffusion timestep t. TAAP is simple, integrating seamlessly into standard training with no changes to sampling or model architecture.

Experiments on CelebA-64 show that TAAP achieves a 21% FID improvement over vanilla DDPM and substantially outperforms the fixed-weight identity loss baseline, which unexpectedly performs worse than vanilla DDPM. The learned weight schedule reveals that identity preservation importance varies significantly across timesteps, confirming that adaptive, data-driven supervision is superior to uniform weighting for attribute preservation in diffusion-based face generation.

2 Related Work

Recent works on identity-preserving face generation have explored various approaches to maintain facial identity consistency during diffusion-based synthesis. ID^3 [7] addresses identity preservation through adjusted likelihood formulations with modified sampling procedures. IPDM [10] applies explicit identity conditioning with uniform supervision across all timesteps, while InstantID [12] introduces identity adapters for inference. Timestep-dependent weighting strategies have also been explored: Min-SNR Weighting [2] uses SNR-based weights for training efficiency, SpeeD [11] employs adaptive sampling for computational acceleration, and Adaptive Non-uniform Timestep Sampling [6] learns which timesteps to prioritize during training. Unlike other works, this work investigates learned per-timestep importance weights for identity preservation via a lightweight auxiliary network, integrating seamlessly into standard DDPM training with minimal computational overhead.

3 Method

3.1 Background: Denoising Diffusion Probabilistic Models

Denoising Diffusion Probabilistic Models (DDPMs) define a forward diffusion process that gradually adds Gaussian noise to an image x_0 over T timesteps, producing increasingly noisy versions $x_1, x_2, \ldots, x_T$. The forward process is defined as:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t;\ \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{3}$$

where β_t is a variance schedule. Using the reparameterization trick, I can sample x_t directly from x_0:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

where $\bar{\alpha}t = \prod i = 1^t(1-\beta_i)$. The model $\epsilon_\theta(x_t, t)$ learns to predict the noise ϵ added at timestep t. Standard DDPM training minimizes:

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}x_0, \epsilon, t\left[|\epsilon - \epsilon_\theta(x_t, t)|^2\right] \tag{5}$$

3.2 Timestep-Adaptive Attribute Preservation (TAAP)

TAAP augments standard DDPM training with an adaptive identity-preserving loss. The key innovation is learning per-timestep importance weights that determine how much emphasis to place on identity preservation at each diffusion step.

Timestep Attribute Weighter Module. I introduce a lightweight neural network $W_\phi(t)$ that maps each timestep $t \in 1, \ldots, T$ to a learned importance weight $\alpha_t \in [0 : 1]$:

$$\alpha_t = W_\phi(t) = \sigma(\text{MLP}_\phi(e_t)) \tag{6}$$

where e_t is the timestep embedding (obtained using sinusoidal positional encoding), MLP_ϕ is a multi-layer perceptron with parameters ϕ, and σ is the sigmoid activation ensuring $\alpha_t \in$. The architecture consists of:

- Embedding layer: $e_t \in \mathbb{R}^{32}$ (sinusoidal encoding of t)
- Hidden layer: Linear(32, 64) + ReLU
- Output layer: Linear(64, 1) + Sigmoid

Identity Loss with Pretrained Model. To measure identity preservation, I employ a frozen pretrained face recognition model f_ψ (InceptionResnetV1 trained on VGGFace2). Given the original image x_0 and the predicted denoised image $\hat{x}_0^t$, I extract identity embeddings:

$$z_{\text{real}} = f_\psi(x_0), \quad z_{\text{gen}} = f_\psi(\hat{x}_0^t) \tag{7}$$

where $\hat{x}_0^t$ is reconstructed from the noisy image x_t using the predicted noise:

$$\hat{x}_0^t = \frac{x_t - \sqrt{1 - \bar{\alpha}t}\epsilon\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{8}$$

The identity loss is computed as:

$$\mathcal{L}_{\text{identity}} = 1 - \text{cosSim}(z_{\text{real}}, z_{\text{gen}}) \tag{9}$$

Note that z_{real} is computed with gradients disabled to avoid backpropagating through the pretrained model. The identity encoder f_ψ remains frozen throughout training.

Adaptive Loss Formulation. The total training loss combines DDPM noise prediction loss with timestep-weighted identity loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DDPM}} + \alpha_t \cdot \mathcal{L}_{\text{identity}} \tag{10}$$

where $\alpha_t = W_\phi(t)$ is the learned timestep-specific weight. Both the diffusion model ϵ_θ and the weighter W_ϕ are trained jointly using gradient descent.

4 Evaluation

4.1 Dataset and Experimental Setup

I evaluate TAAP on CelebA-64, a widely-used benchmark for face generation tasks[8]. CelebA is a large-scale face attributes dataset containing over 200,000 celebrity face images. To reduce computational costs while maintaining sufficient data diversity, I use a random subset

of 30,000 images. All images are center-cropped and resized to 64×64 pixels to maintain sufficient resolution for identity preservation evaluation. Following the standard CelebA data split, the training set comprises 27,000 images and the test set comprises 3,000 images

I compare three baselines:

1. **Vanilla DDPM:** Standard DDPM with only noise prediction loss (Equation 3), without any identity preservation mechanism.
2. **Fixed-Weight Identity Loss:** DDPM with constant identity loss weight ($\lambda = 0.5$), applying uniform supervision across all diffusion timesteps.
3. **TAAP:** My proposed method with learned adaptive per-timestep weights (Equation 5).

All baselines share identical U-Net architectures, training procedures, hyperparameters, and random seed to ensure fair comparison. The only difference is the loss weighting scheme.

4.2 Evaluation Metrics

Both quantitative and qualitative methods are used to assess image quality:

- **Visual Assessment:** Generated samples from all models are compared for perceptual quality and identity.
- **FID (Fréchet Inception Distance)[3]:** Measures similarity between real and generated image distributions, with lower FID indicating better quality.
- **LPIPS[14]:** Evaluates perceptual similarity, where lower values correspond to higher visual fidelity.

5 Results

Figure 1: Samples generated by Vanilla DDPM (**Top**), Fixed-Weight Identity Loss (**Middle**) and TAAP (**Bottom**)

Method	FID ↓	LPIPS ↓
Vanilla DDPM	257.7	0.428
Fixed-Weight Identity Loss($\lambda = 0.5$)	410.5	0.444
TAAP (Mine)	**203.9**	**0.383**

Table 1: Quantitative comparison of TAAP with baseline methods on CelebA-64. Lower FID and LPIPS indicate better generation quality. Best results are shown in bold.

5.1 Analysis of Qualitative Results

Figure 1 shows generated samples for randomly selected identities from TAAP and two baselines. The Vanilla DDPM model produces faces with somewhat recognizable shapes but blurred features, poor facial coherence, and significant color artifacts. Facial structures appear fragmented and lack clear identity. The Fixed-Weight Identity Loss baseline produces almost unrecognizable images, with heavily distorted, uniform, and blocky appearances lacking distinct facial features; outputs appear very similar with no clear identities. In contrast, TAAP generates faces that are much clearer, with well-defined facial features and less noise. Identities are more distinguishable and images look more natural. Notably, TAAP's samples tend to be brighter, while both baseline models produce darker, duller images, suggesting adaptive weighting better preserves relevant face information.

5.2 Analysis of Quantitative Results

Generation Quality (FID). Vanilla DDPM achieves an FID of 257.7, representing poor unconditional generation quality. Surprisingly, Fixed-Weight Identity Loss performs worse with an FID of 410.5, suggesting that naive uniform weighting of identity loss can harm generation quality by over-constraining the model. TAAP achieves a significantly better FID of 203.9, a 21.0

Perceptual Quality (LPIPS). Vanilla DDPM has an LPIPS of 0.428. Fixed-Weight achieves an LPIPS of 0.444, worse than Vanilla DDPM, confirming fixed uniform weighting degrades perceptual quality. TAAP achieves 0.383, representing a 10.5

6 Future Work

Despite TAAP's promising results, several limitations remain.

Dataset Size and Diversity. Due to GPU memory and runtime limits on Google Colab, training used a subset of 30,000 images from the full CelebA dataset (200,000 images). This reduced size may limit generalizability; future work should train on full CelebA or larger datasets such as FFHQ[5] or CelebA-HQ[13].

Hyperparameter Optimization. Default learning rates and hyperparameters were used due to resource constraints. Systematic hyperparameter tuning could improve model performance and stability.

Method Extensions. TAAP could be extended to adaptive weighting for other semantic attributes like expression, pose, or lighting, and integrated into other diffusion variants, such as latent or conditional diffusion, to improve flexibility.

Inference Speed Optimizations. Although TAAP adds no inference overhead compared to vanilla DDPM, combining adaptive weighting with accelerated samplers (e.g. DPM-Solver[9]) could reduce generation time for practical deployment.

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A Appendix: Additional Details

A.1 Implementation Details

Model Architecture. I use a standard U-Net architecture with channel multipliers $(1, 2, 4)$, 2 residual blocks per level, and attention at 32×32 resolution. The model has approximately 45M parameters.

Identity Encoder. I employ InceptionResnetV1 pretrained on VGGFace2 as the frozen identity encoder f_ψ. Input images are resized to 160×160 and normalized to $[-1, 1]$ before passing through the encoder, which outputs 512-dimensional identity embeddings.

Training Configuration. I train all the model for 50 epochs on CelebA-64 with batch size 64, learning rate 2×10^{-4} for the DDPM model and 1×10^{-4} for the Timestep Weighter, using AdamW optimizer with weight decay 1×10^{-4}. Gradient clipping with max norm 1.0 is applied. The noise schedule is linear with $\beta_1 = 0.0001$ and $\beta_T = 0.02$ over $T = 1000$ timesteps.

Sampling. At inference, I use DDIM sampling with 50 steps for faster generation. The Timestep Weighter is not used during sampling—only the trained DDPM model ϵ_θ is required, making TAAP fully compatible with standard diffusion sampling algorithms.

A.2 Training Algorithm

Algorithm 1 describes the complete TAAP training procedure.

Algorithm 1 TAAP Training

Require: Dataset $\mathcal{D}$, DDPM model ϵ_θ, Timestep Weighter W_ϕ, Frozen Identity Encoder f_ψ
Require: linear noise schedule with $\beta_1 = 0.0001$ and $\beta_T = 0.02$ over $T = 1000$ timesteps
  Initialize $\epsilon\theta$ and W_ϕ with random weights
  Load pretrained f_ψ and freeze parameters
  for epoch = 1 to N_{epochs} **do**
    for each batch $x_0^{(i)} i = 1^B \sim \mathcal{D}$ **do**
      Sample $t \sim \text{Uniform}(1, T)$ for each image
      Sample $\epsilon \sim \mathcal{N}(0, I)$
      Compute noisy images: $x_t = \sqrt{\bar{\alpha}t} x_0 + \sqrt{1 - \bar{\alpha}t}\epsilon$
      Predict noise: $\epsilon_{\text{pred}} = \epsilon\theta(x_t, t)$
      Compute DDPM loss: $\mathcal{L}_{\text{DDPM}} = |\epsilon - \epsilon_{\text{pred}}|^2$
      // Identity preservation with frozen pretrained model
      Reconstruct: $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}t}\epsilon_{\text{pred}}) / \sqrt{\bar{\alpha}t}$
      Extract identity embeddings: $z_{\text{real}} = f_\psi(x_0)$, $z_{\text{gen}} = f_\psi(\hat{x}0)$
      Compute identity loss: $\mathcal{L}_{\text{identity}} = 1 - \text{cosSim}(z_{\text{real}}, z_{\text{gen}})$
      // Adaptive weighting
      Compute timestep weight: $\alpha_t = W_\phi(t)$
      Total loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DDPM}} + \alpha_t \cdot \mathcal{L}_{\text{identity}}$
      Update θ and ϕ via SDG
    end for
  end for

The key advantage of this architecture is modularity: the pretrained identity encoder f_ψ remains frozen, the Timestep Weighter W_ϕ is a lightweight addition (only 5K parameters), and the DDPM model ϵ_θ retains its standard architecture.